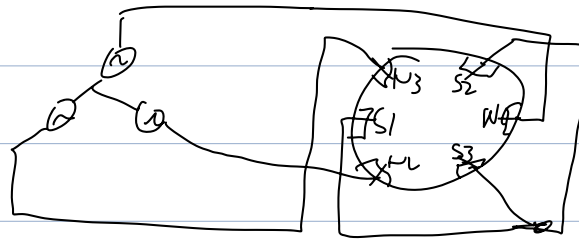
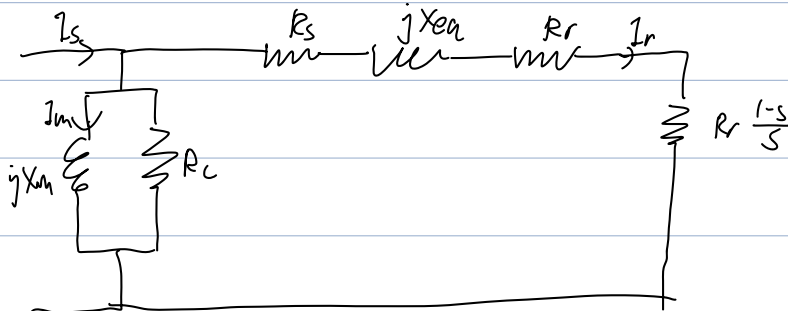
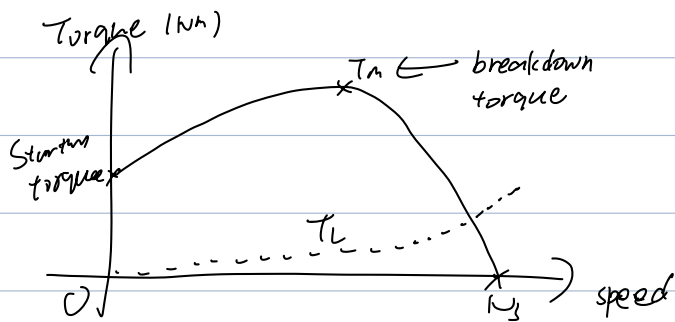


Star connected 3Phase

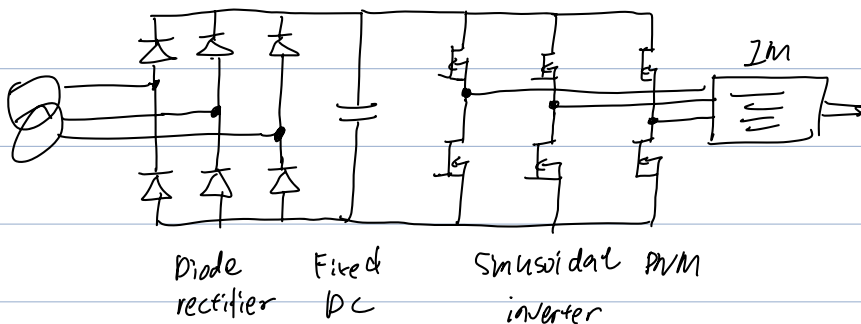
Induction motor



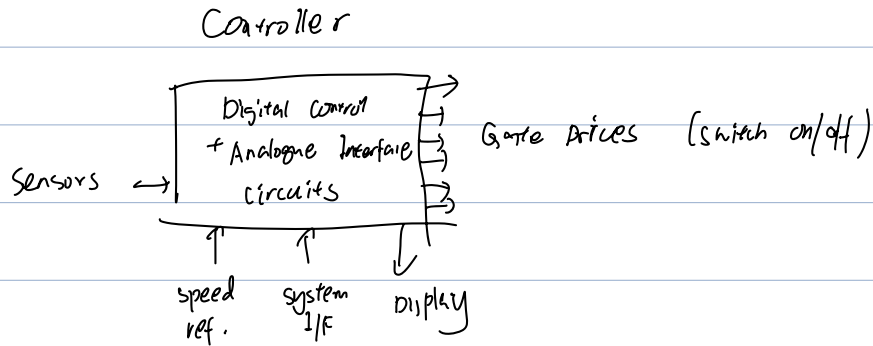
$$N_s = \frac{120 \times f_s}{P} \quad P: \text{number of poles.}$$



Variable speed? driver:



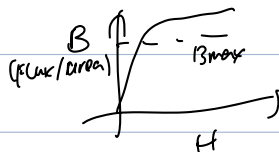
+



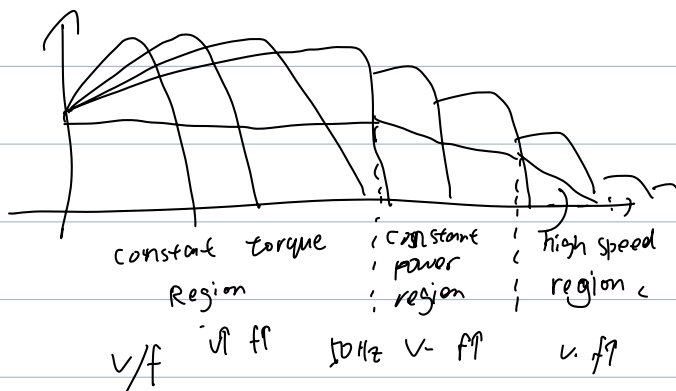
How to control N_s ? method 1: f , since $N_s = \frac{120f}{P}$
 2: V ,
3: Sin AC load current.

We need to keep V/Hz a constant.
 to avoid magnetic flux saturation.

why: Faraday's law: $V = N \cdot \frac{d\phi}{dt} \Rightarrow \phi = \frac{1}{N} \int V dt$

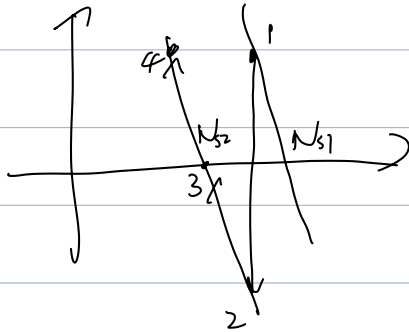


T vs speed for const. V/Hz



$$T_{max} \propto \frac{V}{\omega_s} \propto \frac{V}{f}$$

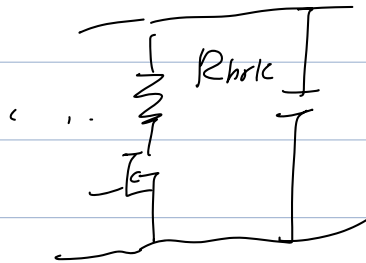
在变速过程中:



2 → 3. returned to the
inverter DC Link

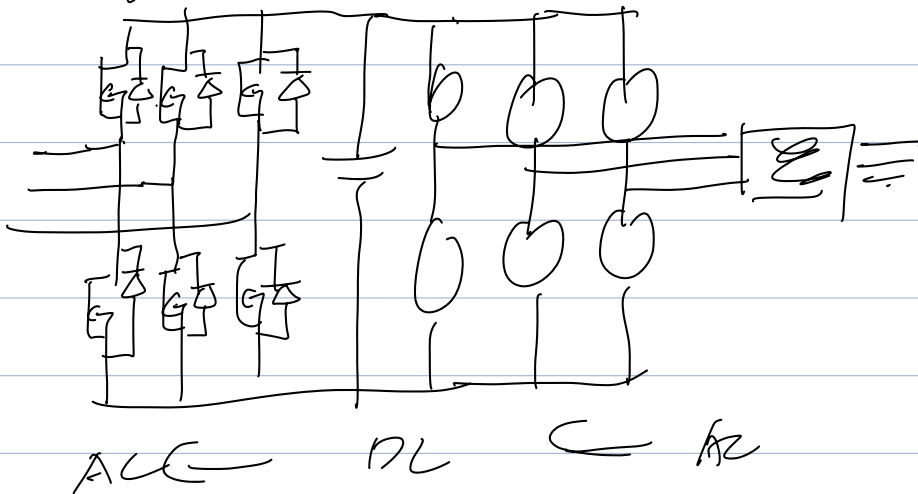
How to avoid DC Link voltage rises:

①



switch controlled
DC Link Braking Resistor

② Regenerative Braking.



e.g. $P_{\text{mech}} = 60 \text{ kW}$

$$P_{\text{brk}} = 2 P_{\text{mech}} = 120 \text{ kW}$$

$$E_{880} = \frac{1}{2} C V_{dc}^2 = 19198 \text{ J}$$

$$E_{900} = \frac{1}{2} C V_{dc}^2 = 19035 \text{ J}$$

$$\Delta E = E_{900} - E_{880} = 837 \text{ J}$$

$$t_{\text{eff}} = \frac{\Delta E}{P_{\text{mech}}} = 13.9 \text{ ms} \quad \uparrow$$

$$P_{dc} = |P_{\text{mech}} - P_{\text{brake}}| = 60 \text{ kW}$$

$$t_{\text{on}} = \frac{\Delta E}{P_{dc}} = 13.9 \text{ ms}$$

$$R_{\text{brk}} = \frac{V_{dc}^2}{P_{\text{brake}}} = 6.6 \Omega.$$

$$T = t_{\text{on}} + t_{\text{eff}} = 27.9 \text{ ms}$$

$$f = \frac{1}{T} = 35.8 \text{ Hz}.$$